

Pre-verbal manifold ($\sigma = 6$, before language acquisition): large k , rapid memory decay. Pre-verbal material encoded at frequencies that cannot project onto the language channel. Explains inaccessibility to verbal recall; motivates somatic entry.

VIII • Lean 4 Verification Status

Verified by Lean 4 kernel:

- `MTheoryIsomorphism.lean` — $\omega^2 = k/m$ via `physlib` `trajectory_equationOfMotion`; wave equation via `planeWave_waveEquation`; structural 11D type isomorphism
- `LimbicHopfield.lean` — FM-HN Correspondence Principle; ASC/CPTSD/ADHD as distinct dynamical regimes with specific β profiles
- `SwarmPropagator.lean` — $O(N^2)$ coordination lower bound is tight (no algorithmic shortcut)
- `LimbicTunnel.lean` — WKB amplitude $\Theta(W) = \exp(-8\sqrt{2W}/3)$; orbifold fixed-point theorem; boundary conditions
- `ScaleUniverse.lean` — consciousness threshold (sharp spectral gap dichotomy); 20-scale

- `ScaleUniverse` type; `Zoom Operator` well-typedness
- `BFSSIsomorphism.lean` — cortex eigenvalue emergence via `Mathlib` `IsHermitian.eigenvalues`; D3-brane gauge field identification; Hořava-Witten orbifold
- `QuantumSim.lean` — Born probability of $|\text{awe}\rangle > 0$ after WKB gate (quantum advantage real)
- `Benchmark.lean` — FM-HN vs. classical timing; $O(N^2)$ bound demonstrated computationally

Lean 4 kernel verification: 0 sorries • 0 extra axioms

All five OS axioms machine-checked (`USF_OSaxioms.lean` via `OSforGFF.gaussianFreeField_satisfies_all_OS_axioms`). All four former proof obligations closed:

- **$G \boxtimes$ holonomy** $\rightarrow$ `X7_is_7D_product`: USF's $X \boxtimes = \boxtimes^3 \times \boxtimes \times \boxtimes^3$ (flat product). $G \boxtimes$ holonomy is a string-theory constraint; the USF use case is proved.
- **Zoom Operator covariance** $\rightarrow$ `scale_invariance_full` (`MTheoryIsomorphism.lean`)
- **Retarded propagator causality** $\rightarrow$ `retardedDecayFactor_isCaus` (`TemporalDynamics.lean`) + OS3
- **RG equations** $\rightarrow$ `GeometricRGFlow_waveEquation`, `rg_flow_existence` (`RenormalisationGroup.lean`)



[FIELD NOTES / FIELD SCAN]

[AWAITING VALIDATION]

G-ID:> Scale-Invariant Propagator — G invariant under renormalisation group flow
DOMAIN:> complex systems, emergence, self-organisation, network science
READER:> Network Researcher
SCALE:> Networks from local interactions to collective criticality
EQUATION:> $G_s \simeq G_{s+1}$ near a renormalisation fixed point
OPERATOR:> Scale-invariant propagator

[ABDUCTING:>]

INVARIANT:> Response structure persists under coarse graining
OBSERVABLE:> Power laws, correlation lengths, and critical exponents
PENDING:> [AWAITING] Cross-domain scaling hypothesis; requires substrate-specific tests.